



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 L1

Fall Semester 2023

Instructor: Usman Salahuddin

Hours: Tues – Thurs

(3:30 PM – 4:45 PM)

Class Location W-242

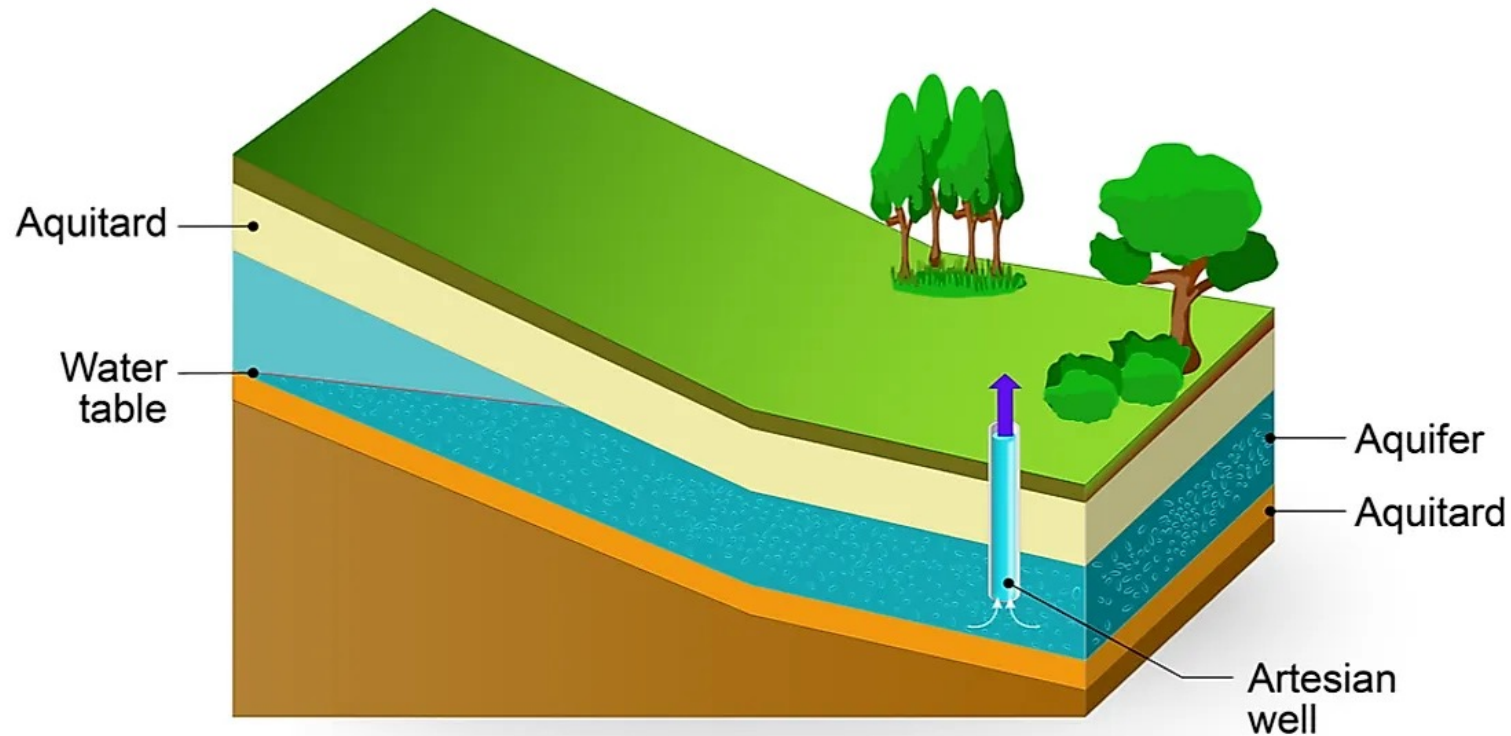
Office hour: Wed 2-3PM

Office: C-103

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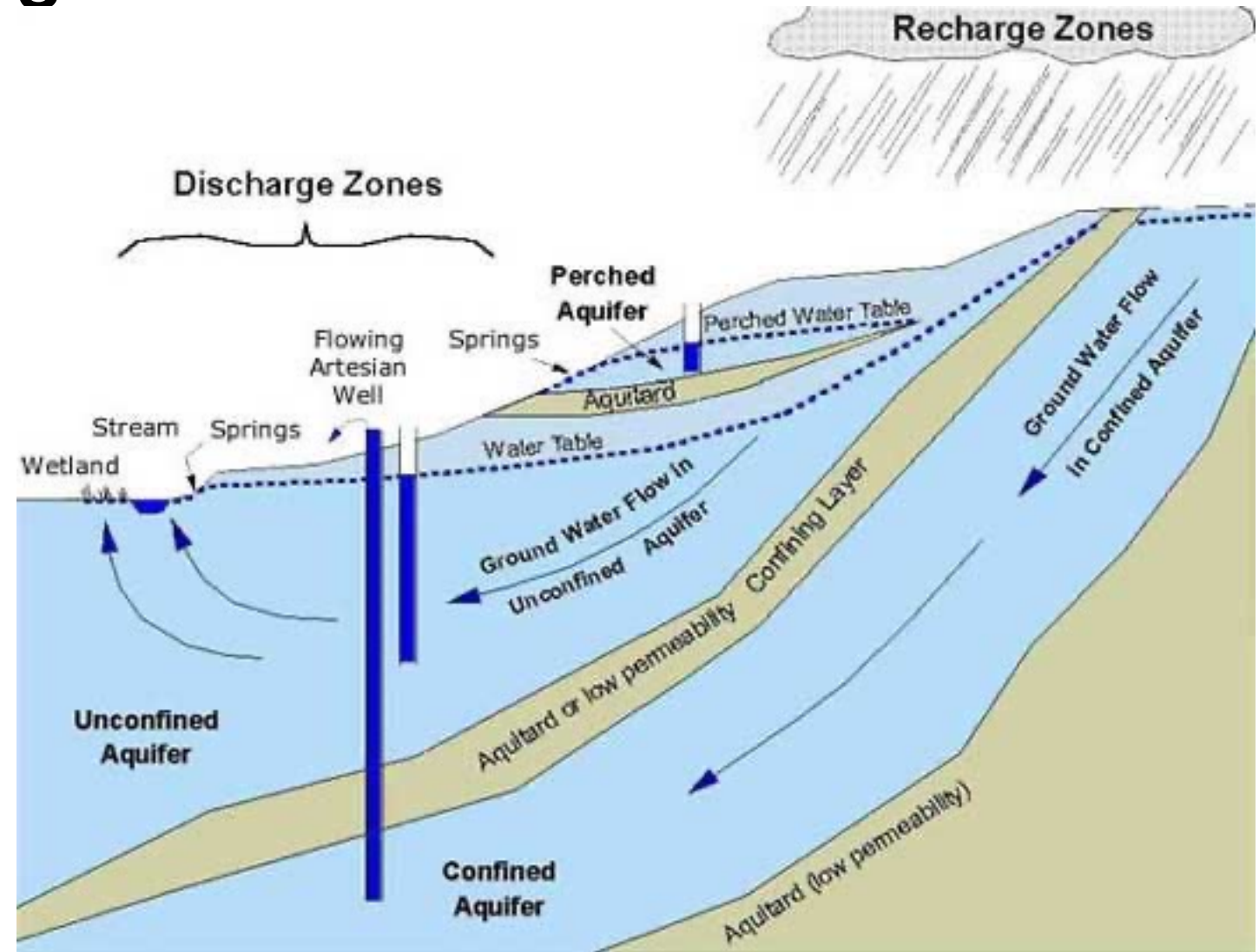
Aquifers

A body of porous rock or sediment saturated with groundwater



Types of Aquifers

- Confined Aquifers
- Unconfined Aquifers
- Perched Aquifers



Groundwater formations

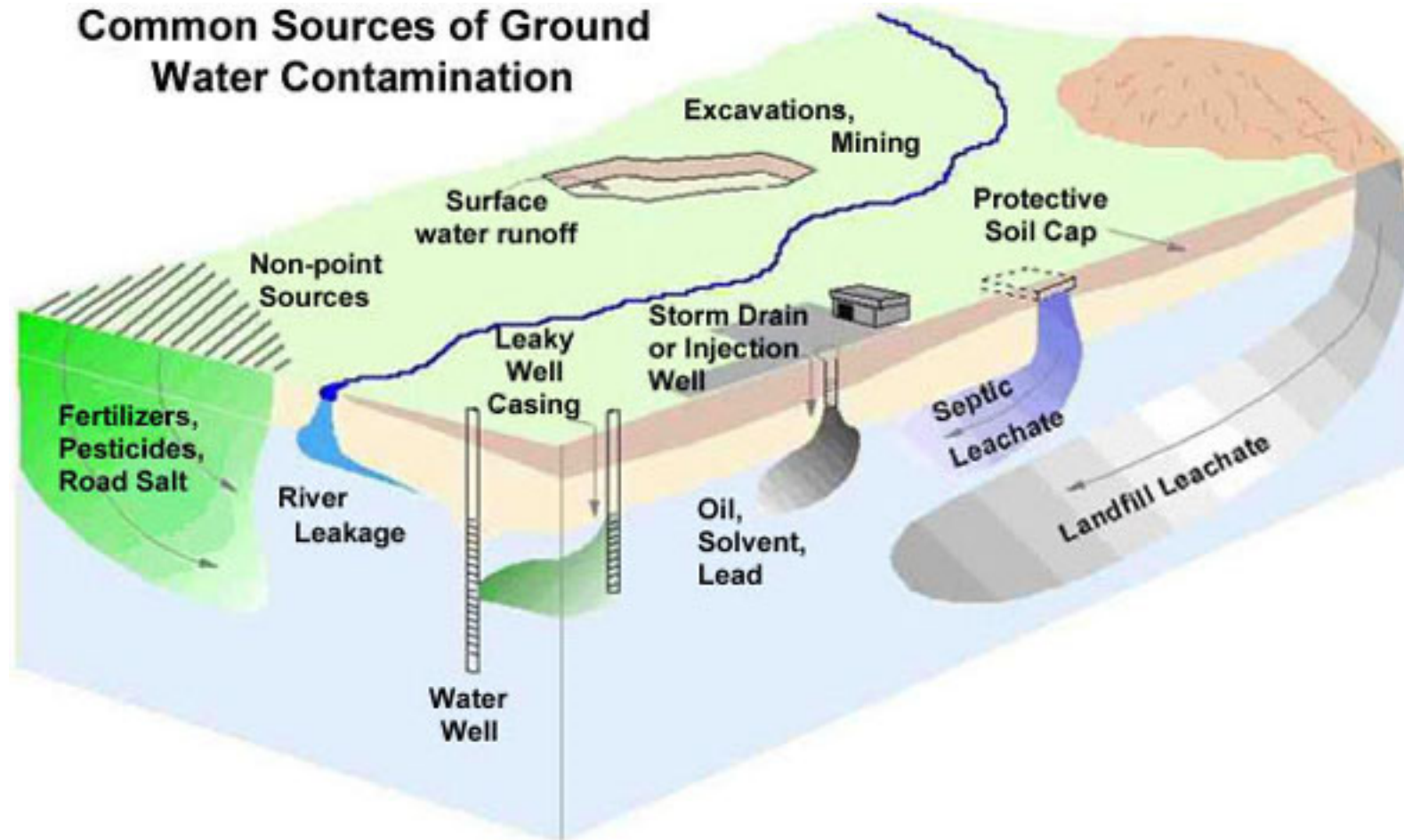
Geological formations/ Properties	Aquifer	Aquitard	Aquiclude	Aquifuge
Water storage	Yes	Yes	Yes	No
Permeability of water	Permeable	Partly permeable	Impermeable	Impermeable
Yield of water	Yes	Yes but slow yielding	Do not yield	Do not yield
Examples	Sand, Gravel	Sandy clay	Clay	Compact rocks such as granite, basalt etc.

Clean Groundwater

Aquifers are natural filters that trap sediment and other particles (like bacteria) and provide natural purification of the ground water flowing through them



Groundwater Contamination



Overview of common aquifer materials:

- Sand
- Gravel
- Sandstone
- Limestone
- Clay



Activity (Groundwater movement)

Link: <https://has.concord.org/groundwater-movement.html>

